

Adding Swap Partitions

Using swap on Linux is a convenient way to improve Linux kernel memory usage. If a shortage of physical RAM occurs, non-recently-used memory pages can be moved to swap, which makes more RAM available for programs that need access to memory pages. Most Linux servers for that reason are configured with a certain amount of swap. If swap starts being used intensively, you could be in trouble though, and that is why swap usage should be closely monitored .

Creating a Swap Partition

1. Use **fdisk /dev/vda** to open your disk in fdisk. (Use **gdisk** if you are using a disk with a GUID partition table.)
2. Press **n** to add a new partition.
3. Type **t** to change the partition type. If you are using fdisk, use partition type 82. If you are using gdisk, use partition type 8200.
4. Use **mkswap** to format the partition as swap space. Use, for instance, **mkswap /dev/vda6** if the partition you have just created is /dev/vda6.
5. Type **free -m**. You see the amount of swap space that is currently allocated.
6. Use **swapon** to switch on the newly allocated swap space. If, for instance, the swap device you have just created is /dev/vda6, use **swapon /dev/vda6** to activate the swap space.
7. Type **free -m** again. You see that the new swap space has been added to your server.

Mounting File Systems

Mounting Means make FileSystem accessible to end user so he can write data and retrieve this data again.

We used earlier the command mount as **#mount /dev/sdb1 /mnt**

you make its(/dev/sdb1) contents accessible through a specific directory(/mnt).

To mount a file system, some information is needed:

- **What to mount:** This information is mandatory and specifies the name of the device that needs to be mounted.
- **Where to mount it:** This is also mandatory information which specifies the directory on which the device should be mounted.
- **What file system to mount:** Optionally, you can specify the file system type. In most cases, this is not necessary. The **mount** command will detect which file system is used on the device and make sure the correct driver is used.
- **Mount options:** Many mount options can be used when mounting a device. Using options is optional and depends on the needs you may have with the file system.

What to mount ?

You can use one of the following options to refer to the partition you want to mount:

1. Device name (/dev/sdc1, /dev/sdb5,...)

2. Label : Label is just name for you as you can rename a partition under windows with name songs or movies .

To create label for partition :

In ext4 filesystem the command is **e2label**

```
#e2label /dev/sda3 songs
```

In xfs filesystem the command is **xfs_admin -L**

```
#xfs_admin -L songs /dev/sda3
```

3. UUID : every object in your system has it's unique ID. Every Partition you created and applied FileSystem on it the system created UUID for it. (THE BEST WAY).

To display UUID for the partitions in the system use the command **blkid**

To display block devices in the system use command **lsblk** (with -f option to list labels)

Examples

1. mount UUID="42f419c4-633f-4ed7-b161-519a4dadd3da" /mnt
2. mount /dev/sdc4 /media
3. mount LABEL=songs /mnt

Permanent Mount with fstab:

Normally, you do not want to be mounting file systems manually. Once you are ready, it is a good idea to have them mounted automatically. The classical way to do this is through the /etc/fstab file.

```
[root@server3 ~]# cat /etc/fstab
```

```
#
```

```
# /etc/fstab
```

```
# Created by anaconda on Fri Jan 16 10:28:41 2015
```

```
#
```

```
# Accessible filesystems, by reference, are maintained under '/dev/disk'
```

```
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info
```

```
#
```

```
/dev/mapper/centos-root / xfs defaults 1 1
```

```
UUID=02305166-840d-4f74 / xfs defaults 1 2
```

```
/dev/mapper/centos-swap swap swap defaults 0 0
```

Content of FSTAB file

Field	Description
Device	The device that must be mounted. A device name, UUID, or label can be used.
Mount Point	The directory or kernel interface where the device needs to be mounted.
File System	The file system type.
Mount Options	Mount options.
Dump Support	Use 1 to enable support to backup using the dump utility. This may be necessary for some backup solutions.
Automatic Check	Specifies if the file system should be checked automatically when booting. Use 0 to disable automated check, 1 if this is the root file system and it has to be checked automatically, and 2 for all other file systems that need automatic checking while booting. Network file systems should have this option set to 0.

Mounting Options

Option	Use
auto/ noauto	The file system will [not] be mounted automatically.
acl	Adds support for file system access control lists
user_xattr	Add support for user extended attributes
ro	mounts the file system in read-only mode.
atime / noatime	Disables or enables access time modifications.
noexec / exec	Denies or allows execution of program files from the file system.
_netdev	Use this to mount a network file system. This tells fstab to wait until the network is available before mounting this file system.

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